

ENGR 102 Summer Bridge

Day 16 Instructor Key

Exam 2 Rehearsal: Loops, Lists, Mutation, and Filtering

20 points • approximately 20-25 minutes

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Boolean-operator convention. Python operators appear as [and](#), [or](#), and [not](#). Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace skip, running state, collection, and early stop.

6 points

```
values = [3, -2, 5, 0]
result = []
running = 0

for index in range(len(values)):
    if values[index] < 0:
        continue
    running = running + values[index]
    if running > 6:
        result.append(running)
        break
    result.append(values[index] * 2)

print(result, running, index)
```

Give a complete reached-pass ledger, including skipped statements and the exact final output.

Answer and explanation

Index 0 adds 3 and appends 6. Index 1 is negative, so continue skips both updates. Index 2 adds 5, making running 8; the > 6 branch appends 8 and break ends the loop before index 3. Exact output: [6, 8] 8 2.

Question 2. Diagnose final-list evidence that disagrees with the total.

4 points

```
values = [4, -3, 7, -1]
total = 0

for value in values:
    if value < 0:
        value = 0
    total = total + value

print(values, total)
```

Give actual output, explain the mismatch, and write an indexed repair that makes the list and total describe the same final state.

Answer and explanation

Temporary value becomes 0 on negative passes, so total is 11, but the list remains [4, -3, 7, -1]. Repair with an index loop that assigns values[index] = 0 before adding values[index]. The repaired final list is [4, 0, 7, 0] and total remains 11.

Question 3. Program construction**10 points**

Write one complete Python program that satisfies every requirement.

- Read threshold as an integer and use values = [12, -4, 7, 20, 7].
- Loop by index and replace negative values with 0 before any summary.
- Append a final value to selected only when it is at least threshold and even.
- Compute the total of all final values and the index of the first largest final value without max or index.
- Print Final, Selected, Total, and Largest index on separate labeled lines. Give a threshold-equality test.

Answer and explanation

```
threshold = int(input())
values = [12, -4, 7, 20, 7]
selected = []
total = 0
largest = values[0]
largest_index = 0

for index in range(len(values)):
    if values[index] < 0:
        values[index] = 0
    value = values[index]
    total = total + value
    if value >= threshold and value % 2 == 0:
        selected.append(value)
    if value > largest:
        largest = value
        largest_index = index

print(f"Final: {values}")
print(f"Selected: {selected}")
print(f"Total: {total}")
print(f"Largest index: {largest_index}")
```

All summaries follow mutation, so they describe one consistent final list. The compound filter requires both threshold evidence and evenness. Strict > preserves the first largest position.

With threshold 8, final values are [12, 0, 7, 20, 7], selected is [12, 20], total is 46, and largest_index is 3. Threshold 20 must still select 20 because equality is included.

Scoring guidance

1 input, 2 indexed mutation, 2 compound selection, 1 total, 2 first-largest tracking, 1 exact output, 1 equality test.